

CES Utility: What Changes as s Changes?

Q1(b) Explained

ECO 310

The Big Confusion

When Q1(b) asks "what happens as $s \rightarrow 0$ ", students often think the budget line changes.

THE BUDGET LINE DOES NOT CHANGE!

The budget line is:

$$p_1x_1 + p_2x_2 = m$$

It only depends on **prices and income**, which are fixed in Q1(b).

What DOES Change?

Your indifference curves change shape!

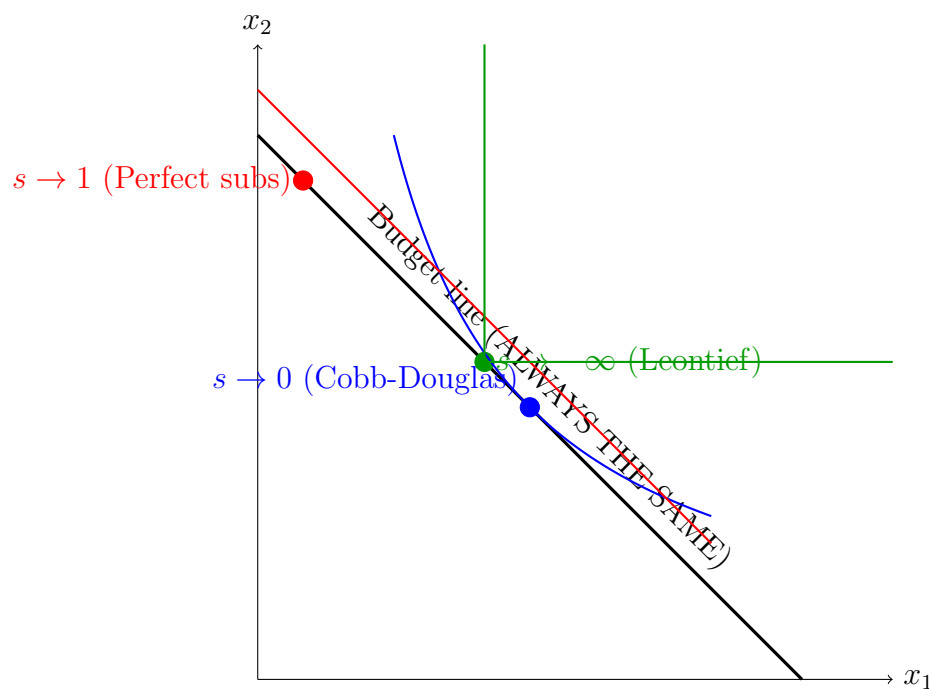
The CES utility function is:

$$u(x_1, x_2) = (x_1^s + x_2^s)^{1/s}$$

As s changes, the shape of your indifference curves changes, which makes you choose **different points on the same budget line**.

Visual Explanation

Imagine a fixed budget line (same prices, same income):



Same budget line, different indifference curves, different optimal choices!

The Three Cases

Case 1: $s \rightarrow 0$ (Cobb-Douglas)

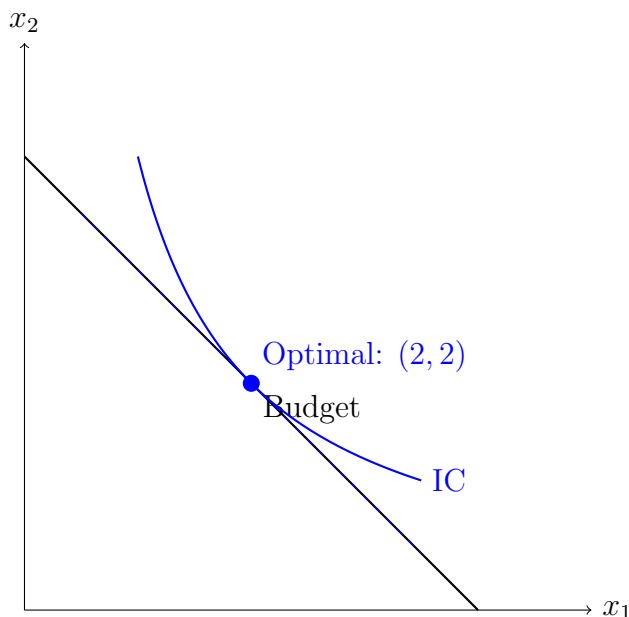
Indifference curves: Smooth, curved hyperbolas

Demand functions approach:

$$x_1^* = \frac{m}{2p_1}, \quad x_2^* = \frac{m}{2p_2}$$

Interpretation: You spend a constant fraction of income on each good (50% on good 1, 50% on good 2).

On the budget line: You choose an **interior point** where the IC is tangent to the budget line.



Key: Interior solution with tangency.

Case 2: $s \rightarrow 1$ (Perfect Substitutes)

Indifference curves: Straight lines with slope $= -1$

Demand functions approach:

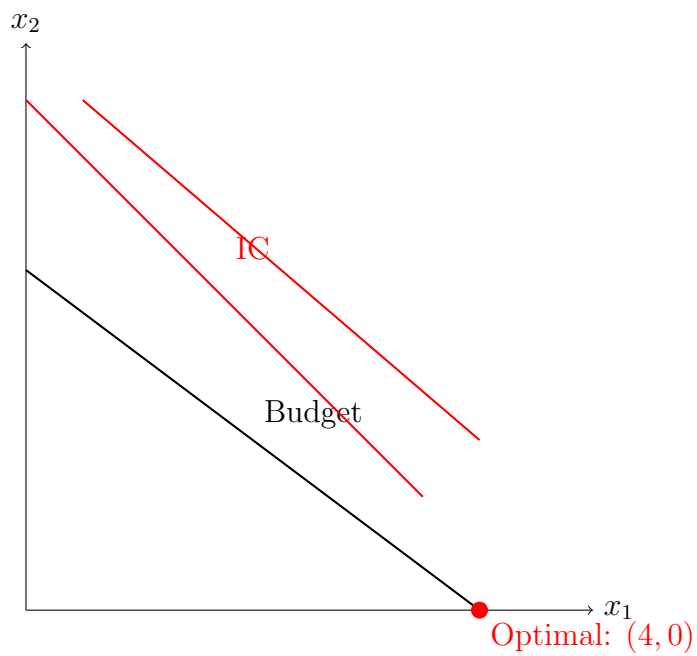
$$\text{If } p_1 < p_2 : \quad x_1^* = \frac{m}{p_1}, \quad x_2^* = 0$$

$$\text{If } p_2 < p_1 : \quad x_1^* = 0, \quad x_2^* = \frac{m}{p_2}$$

$\text{If } p_1 = p_2 : \quad \text{Any bundle on the budget line}$

Interpretation: Buy only the cheaper good (corner solution).

On the budget line: You choose a **corner point** (all of one good, none of the other).



Key: Corner solution - buy all of the cheaper good.

Case 3: $s \rightarrow -\infty$ (Perfect Complements / Leontief)

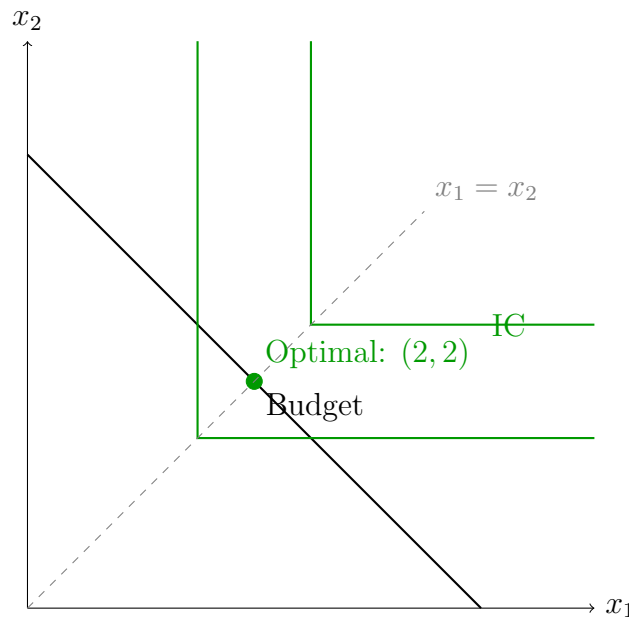
Indifference curves: L-shaped with kink at $x_1 = x_2$

Demand functions approach:

$$x_1^* = \frac{m}{p_1 + p_2}, \quad x_2^* = \frac{m}{p_1 + p_2}$$

Interpretation: You consume goods in a fixed ratio (here, 1:1).

On the budget line: You choose the point where the **kink** of the IC touches the budget line.



Key: Optimal at the kink - consume in fixed ratio.

Summary Table

As $s \rightarrow$	IC Shape	Optimal Point on Budget Line
0 (Cobb-Douglas)	Smooth curves	Interior tangency
1 (Perfect subs)	Straight lines	Corner (buy cheaper good)
$-\infty$ (Leontief)	L-shaped	At the kink (fixed ratio)

The Key Insight for Q1(b)

What changes: The shape of indifference curves

What stays the same: The budget line (same prices, same income)

Result: You choose different bundles on the same budget line

How to Answer Q1(b)

Q1(b) asks: "Show that as s approaches certain values, your demand functions from (a) approach those for [Cobb-Douglas / Perfect Substitutes / Leontief]."

Step 1: Know the Target Demand Functions

Cobb-Douglas $u = x_1^{1/2} x_2^{1/2}$:

$$x_1^* = \frac{m}{2p_1}, \quad x_2^* = \frac{m}{2p_2}$$

Perfect Substitutes $u = x_1 + x_2$:

$$\text{If } p_1 < p_2 : \quad x_1^* = \frac{m}{p_1}, \quad x_2^* = 0$$

$$\text{If } p_2 < p_1 : \quad x_1^* = 0, \quad x_2^* = \frac{m}{p_2}$$

Leontief $u = \min\{x_1, x_2\}$:

$$x_1^* = x_2^* = \frac{m}{p_1 + p_2}$$

Step 2: Take Your CES Demand from (a)

From Q1(a), you should have derived:

$$x_1^* = \frac{m \cdot p_1^{\frac{s}{s-1}}}{p_1^{\frac{s}{s-1}} + p_2^{\frac{s}{s-1}}}$$

(or something similar depending on normalization)

Step 3: Take Limits

For $s \rightarrow 0$:

Use L'Hospital's rule or the fact that as $s \rightarrow 0$:

$$p^{\frac{s}{s-1}} \rightarrow p^{-1}$$

So:

$$x_1^* \rightarrow \frac{m \cdot p_1^{-1}}{p_1^{-1} + p_2^{-1}} = \frac{m/p_1}{1/p_1 + 1/p_2} = \frac{m}{2p_1}$$

(assuming $p_1 = p_2$ for simplicity in the equal-weight case)

This matches Cobb-Douglas!

For $s \rightarrow 1$:

As $s \rightarrow 1$:

$$\frac{s}{s-1} \rightarrow \infty$$

So $p^{\frac{s}{s-1}}$ explodes for the larger p , meaning you spend all on the cheaper good.

This matches perfect substitutes!

For $s \rightarrow -\infty$:

The CES function approaches:

$$u = \min\{x_1, x_2\}$$

With kink condition $x_1 = x_2$ and budget constraint:

$$p_1x_1 + p_2x_2 = m \Rightarrow x_1(p_1 + p_2) = m \Rightarrow x_1 = \frac{m}{p_1 + p_2}$$

This matches Leontief!

What You Write on the Exam

Example for $s \rightarrow 0$:

As $s \rightarrow 0$, the CES utility approaches Cobb-Douglas. Taking the limit of the demand function from (a):

$$\lim_{s \rightarrow 0} x_1^* = \frac{m}{2p_1}$$

This matches the Cobb-Douglas demand where consumers spend a constant fraction of income on each good, independent of prices (in the equal-weight case).

The budget line remains $p_1x_1 + p_2x_2 = m$, but the indifference curves become smooth hyperbolas, leading to an interior tangency solution.

Key points to include:

- State what the limit is
- Show the demand approaches the target form
- Mention the shape of ICs (optional but helpful)
- Note the budget line stays the same (if asked)

Common Mistakes

1. **Saying the budget line changes** - No! Only prices/income change the budget line.
2. **Not taking the actual limit** - You need to show the algebra, not just say "it approaches."
3. **Confusing s values** - $s \rightarrow 0$ is Cobb-Douglas, $s \rightarrow 1$ is perfect subs, $s \rightarrow -\infty$ is Leontief.
4. **Forgetting what changes vs what stays the same** - ICs change shape; budget line stays fixed.